

DA-02-T9

Prototype Embedded Dashboard Insights — Product Comparison Page

DiscountMate | DataBytes | Trimester 2 2026 | Feature 2: Insight Surfacing Power BI Dashboards

Task Overview

Item	Detail
Context	Prototype embedded dashboard insights for the Product Comparison page
Scope	Plan embedded insights; prototype the dashboard integration (visual prototype only — technical embedding is out of scope for this task).
Approach	This is a design/planning prototype. It shows how selected Power BI pricing insights could appear inside the DiscountMate product interface.

References used: Product Comparison design (single-product and grocery-list views), the DA-02-T2 stakeholder views, DA-02-T3 dashboard requirements and user flows, the DA-02-T4 behavioural data requirements documentation, and the DA-02-T8 Segment Profiles & Segment overview.

1. Purpose of the Embedded Insights and Value to Customers

Purpose:

- Surface a small set of high-value pricing insights directly inside the Product Comparison page, instead of requiring customers to leave the app.
- Reuse the pricing calculations that already exist in the Power BI model so that customers see the same numbers Feature 2 already trusts, just presented in-context.
- Support the customer requirements already agreed in DA-02-T3 — in particular C3 (identify the lowest/best price), C4 (price history), C8 (savings and price metrics) and C9 (explain discount measures simply).

Value to customers:

- Faster decisions — the customer can see whether a product is currently a good buy without leaving the comparison screen.
- Confidence — a visible price trend and "lowest price" indicator reduce the worry of buying too early or missing a better deal.
- Transparency — showing savings and discount percentage in plain terms builds trust in the retailer comparison.
- Consistency — reusing the same visual language as the existing Product Comparison design (per DA-02-T3 section 10; keep dashboard design consistent with the DiscountMate Product Comparison experience) means no new UI patterns to learn.

2. Where the Insights Would Appear

Draft mock-ups are suggested. In summary:

- Single Product view — a new "Pricing Insights" panel sits directly beneath the existing "Buy now" recommendation card and above the "Compare retailers" table, so it is visible without scrolling past the core comparison content.
- Grocery List view — a compact savings/trend badge could appear next to each line item, linking through to the same panel for that product (optional secondary placement, lower priority than the single-product panel). This is a small, optional suggestion for the existing Grocery List view rather than a fixed design requirement.

3. Useful Customer Insights

The following insights are proposed for the embedded panel, all derived from data and calculations that already exist in the Power BI model per DA-02-T3:

- Current price (per retailer, and at the selected retailer).
- Previous price (for comparison against the current price).
- Price difference between retailers (e.g. "+\$0.41 vs Aldi").

- Savings — dollar amount and percentage saved vs the most expensive option.
- Discount percentage on the current special, where applicable.
- Price trend — a short visual trend (up/down/flat) over a recent window.
- Lowest price — whether the current price is the lowest recorded, or how it compares to the recent average.
- Special status — whether the product is currently on promotion at a given retailer.

4. Filters and Interactions

- Retailer — filter the panel to one retailer or view all retailers side by side (consistent with the existing retailer tabs/logos in the Product Comparison and Category Specials pages).
- Product — the panel updates for which ever product the customer has selected or searched for in the comparison view.
- Category — filter context carried over from Category Specials-style category selection, for browsing rather than single-product lookup.
- Date range — a short set of preset windows (e.g. 4 weeks / 3 months / 6 months / 1 year), matching the existing Product Price Analysis page's date controls, to control the trend view.
- Include similar products (optional toggle) — extends the comparison to near-equivalent products, consistent with the existing single-product comparison design.

5. User Flow

The diagram below shows how a customer reaches and interacts with the embedded insights, from opening Product Comparison through to acting on an insight.

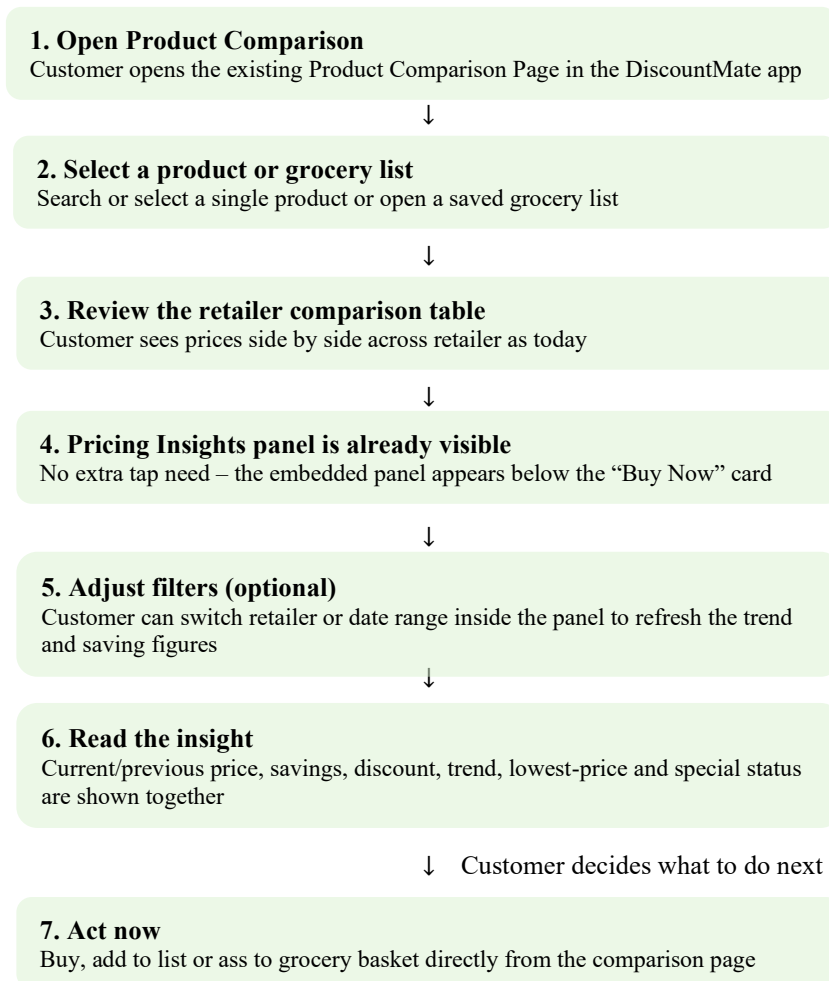


Figure 1: Customer user flow for the embedded Pricing Insights panel.

6. Notes: Data Fields and Power BI Visuals Supporting Each Insight

Insight	Supporting data field(s)	Source	Suggested visual
Current price	Normalised price per retailer	Existing model (DA-02-T3 section 9.1)	Card / table cell (reuse existing Compare Retailers table)
Previous price	Previous price/date calculation	Existing DAX	Card with strike-through, as in current single-product mock-up
Price difference	Retailer price array, difference column	Existing model	Table/matrix visual (reuse existing Compare Retailers table)
Savings (\$ / %)	Savings measure, discount amount	Existing + planned Gold-layer field	KPI card
Discount percentage	Discount percentage field	Existing model	Card / conditional-formatted cell
Price trend	Price-change calculation, previous price/date	Existing model; chart reused from Product Price Analysis	Line chart
Lowest price	Lowest/best price indicator	Existing model	Badge / icon (reuse existing "Lowest Price" indicator from Category Specials)
Special status	Promotion status field	Existing model	Badge (reuse existing "On special" tag)

7. Dependencies and Limitations

- This task delivers a visual prototype only. Actual technical embedding is out of scope and depends on a separate decision with the App Development team on embedding approach.
- The eight pricing insights above are all supported by fields that already exist in the current Power BI model.
- Persona/behavioural insights are explicitly excluded from this prototype. The DA-02-T8 draft dashboard itself notes its results are based on synthetic users pending validation with real GA4 data, reinforcing that this data is not yet production-ready for a customer-facing surface.
- Popular-products and frequently-compared-products insights are also excluded, as DA-02-T4 records these as a future requirement pending product-level event data.
- DA-02-T6 and DA-02-T7 were not yet ready and available during the development of this prototype. Therefore, the prototype could not be fully coordinated or cross-checked against their final outputs and requirements. Alignment with these tasks will need to be confirmed once their relevant outputs become available.
- Design must remain consistent with the existing Product Comparison visual style (DA-02-T3 section 1, C10) and should reuse existing model fields and DAX calculations rather than duplicating them.

Sources: DA-01-T3 — Behavioural Feature Specification (Robin Le); DA-02-T3 — Dashboard Requirements and User Flows; DA-02-T4 — Behavioural Data Requirements for Insight Surfacing; DA-02-T8 — Segment Profiles & overview dashboard; App Development Product Comparison design pages.